



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Information Technology
Academic Year 2024 - 2025 (Even Semester)

Degree, Semester & Branch: B. Tech, VI Semester, IT

Course Code & Title: CS3492 – Database Management Systems

Name of the Faculty Member: A. Alagulakshmi, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit V/ SQL Injection
- **Course Outcome:** CO5
- **Activity Chosen:** Think-Pair-Share (TPS)
- **Justification:**
 - SQL Injection is a critical security flaw in database systems covered in Unit V.
 - This activity was designed to deepen students' understanding of the nature of SQL injection attacks, how they occur, and effective prevention methods.
 - Using the Think-Pair-Share method promotes collaborative learning, encouraging students to think critically, communicate their ideas, and learn from peers.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**

Students were introduced to the topic “SQL Injection: Types, Examples, and Prevention”. The class was organized into 31 pairs, each discussing the topic in detail for five minutes. Pairs exchanged views on different types of SQL injection attacks and brainstormed strategies to prevent such vulnerabilities. Following discussions, selected pairs presented their findings and preventive techniques to the entire class. One exemplary presentation was given by the group led by Mr. Gopi S, who clearly explained SQL injection categories with practical prevention approaches using real-world cases.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO5	PO9	P010	P011	P012	PSO3
CO5	3	2	3	2	2	3	3	3

(1-Low

2-Moderate

3-High)

- **PO / PSO mapped:**

Innovative practice	PO1 3	PO2 2	PO5 3	PO9 2	P010 2	P011 3	P012 3	PSO3 3
Justification for correlation	Applies foundational database and security principles to identify SQL injection threats.	Analyzes different types and impacts of SQL injection attacks.	Understands the role of security tools and best practices for protection.	Promotes teamwork through paired discussions and cooperative problem-solving.	Develops effective oral communication skills through group presentations.	Fosters ethical awareness regarding database security responsibilities.	Encourages continuous learning about emerging security challenges.	Enhances analytical skills by assessing risks and proposing practical solutions in database environments.

- **Images / Screenshot of the practice:**



Figure 1. Presentation of students

Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- Students showed enthusiasm and engaged well in the Think-Pair-Share format.
- Many students appreciated the peer discussions, stating it helped them clarify concepts and retain knowledge better.

❖ Benefit of the practice:

- Enhanced understanding of SQL injection vulnerabilities and mitigation techniques.
- Increased interaction between students and the instructor, as well as among peers.
- Reduced anxiety in sharing ideas, as small group discussions felt less intimidating than large group presentations.

- Encouraged individual reflection coupled with collaborative learning to deepen comprehension.
- Improved communication and teamwork skills related to database security topics.
- ❖ ***Challenges faced in implementation:***
 - Some students were initially reluctant to present in front of the class.
 - Time management was a concern as the activity extended beyond the scheduled 10 minutes due to students needing extra time to build confidence.

References:

- <https://teachingcenter.wustl.edu/resources/active-learning/think-pair-share/>
- <https://teaching.cornell.edu/teaching-resources/engaging-students/active-learning/think-pair-share>
- <https://crlt.umich.edu/tstrategies/ts/thinkpairs>
- <https://www.learningoutcomeassessment.org/Strategies/Think-Pair-Share>

Signature of Faculty Member

HOD